

Deepak Mallampati

APPLIED ML & AI ENGINEER

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | Kent, OH

PROFESSIONAL SUMMARY

Applied ML / AI Engineer with 4+ years shipping production machine learning and LLM systems for regulated and enterprise environments. Built supervised forecasting, anomaly-detection, and classification models on operational sensor data, then scaled into production LLM applications and MLOps - REST APIs, model monitoring, and RAGAS/BERTScore evaluation frameworks. Track record of measurable production outcomes: 42% token reduction on agentic workflows, p95 latency 1.5s → 800ms across 100K+ daily requests, and 60% fewer failed pipeline runs. Strong on responsible-AI governance (content safety, HIPAA/SOC 2) and the AWS / Databricks / MLflow stack.

CORE COMPETENCIES

Time-Series Forecasting • Anomaly Detection • Classification & Regression • LLM Applications • MLOps & Model Monitoring • REST API Development • AWS / Databricks / MLflow • Evaluation Frameworks • Responsible AI Governance

Machine Learning & NLP: PyTorch, TensorFlow, Keras | Forecasting, Anomaly Detection, Classification, Regression | NER, Text Classification, Summarization | Model Optimization, A/B Testing, Drift Detection | RAGAS, BERTScore

MLOps & Cloud: AWS (SageMaker, Bedrock, EC2, S3, Lambda), Databricks, Azure, GCP Vertex AI | Docker, Kubernetes, Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring & Observability

Generative AI & LLMs: GPT-4o/5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety

Data & Retrieval: PostgreSQL, MongoDB, Redis | Spark, Kafka | Pandas, NumPy | FAISS, Pinecone, Weaviate | Data Lineage & Governance

Languages & Tools: Python, Java, JavaScript/TypeScript, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Developed and enhanced **AI agents and backend services** for the internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** on internal evaluation datasets and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Developed and trained supervised **regression and classification models** on operational pipeline-storage data, applying standard ML workflows to **equipment-failure prediction, anomaly detection, and capacity forecasting**.
- Engineered features and executed end-to-end data preprocessing - cleaning, normalization, and transformation of raw sensor and operational datasets - to produce model-ready training pipelines.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific datasets, achieving **89% F1-score** on custom NER tasks.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs. 0.696).
- Designed a composite **privacy-utility scoring framework** benchmarking six anonymization configurations to identify the optimal operating point.
- Authored a peer-review-ready **IEEE-format conference paper** documenting methodology, experiments, and results.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

ML Intern

- Built an end-to-end **video summarization and highlight-generation pipeline** integrating speech-to-text, transformer-based abstractive summarization, and timestamp-aligned clip extraction.
- Deployed across **5,000+ videos**, automating highlight selection and reducing manual editing effort by **70%**.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

A framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy - quantifying exactly how much privacy each technique buys for how little accuracy cost.

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision, compressing hours of repetitive work into an automated workflow.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across 29+ sites, turning a multi-step manual translation chore into a single inline action.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0